Circuit Lab, Division B

Practice Test 1 (Regional level) · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

Instructions

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

Section	Questions
Section I: Multiple choice	1–23
Section II: Short answer and calculations	24–42
Section III: Circuit problems	43–50

Section I: Multiple choice

Circle the one best answer. 1 point each.

1. The SI unit of electric current is named after which scientist?

A. André-Marie Ampère
B. Alessandro Volta
C. Georg Ohm
D. Charles-Augustin de Coulomb

2. Who built the first electric battery, the voltaic pile, in about 1800?

A. Georg Ohm
B. Alessandro Volta
C. Heinrich Hertz
D. Gustav Kirchhoff

3. George Boole is best known for which work?

A. Writing an algebra of logic in which values are true or false
B. Measuring the force between two charges with a torsion balance
C. Proving that electromagnetic waves exist
D. Stating the law V = IR

4. The unit hertz (Hz) measures which quantity?

A. Charge per second
B. Energy per second
C. Volts per meter
D. Cycles per second

5. A rubber balloon is rubbed on dry hair. The balloon becomes negative. What moved?

A. Protons moved from the balloon to the hair
B. Electrons moved from the hair to the balloon
C. Protons moved from the hair to the balloon
D. Neutrons moved from the hair to the balloon

6. A positive rod is held near a neutral metal sphere (it never touches). The sphere is touched to ground for a moment, then the ground is removed, then the rod is taken away. What is the final charge on the sphere?

A. Positive
B. Zero
C. It depends on the size of the rod
D. Negative

7. Two point charges are moved so that the distance between them is 3 times as large. The new force is how many times the old force?

A. 3
B. 1/3
C. 1/9
D. 9

8. Why does a fuel truck connect a grounding cable before it pumps fuel?

A. To charge the fuel so that it flows faster
B. To measure how much fuel is pumped
C. To keep the pump motor cool
D. To stop a static spark that could ignite fuel vapor

9. A standard US household outlet supplies about:

A. 120 V DC
B. 120 V AC at 60 Hz
C. 240 V AC at 50 Hz
D. 12 V AC at 60 Hz

10. On a polarized US plug, one blade is wider than the other. The wide blade connects to the:

A. Neutral wire
B. Hot wire
C. Ground wire
D. Fuse

11. A copper wire is replaced by a copper wire of the same length with TWICE the diameter. The new resistance is:

A. One quarter as large
B. Twice as large
C. Half as large
D. Four times as large

12. Three bulbs are in SERIES with a battery. One bulb burns out (its filament breaks). What happens to the other two bulbs?

A. They get brighter
B. They stay the same
C. One goes out and one gets brighter
D. They go out

13. How is an ammeter connected to measure the current in a resistor, and what is its ideal resistance?

A. In parallel; zero resistance
B. In series; zero resistance
C. In series; very high resistance
D. In parallel; very high resistance

14. Which band colors mark a 680 Ω, ±5% resistor?

A. Blue, gray, red, gold
B. Blue, gray, brown, gold
C. Violet, gray, brown, silver
D. Gray, blue, brown, gold

15. Where is the fuse placed in a household circuit?

A. In series with the hot wire
B. In series with the neutral wire
C. In parallel with the load
D. Between the neutral and ground wires

16. A GFCI outlet trips when:

A. The hot and neutral currents differ by about 5 mA
B. The current goes above 15 A
C. The voltage drops below 100 V
D. The outlet gets too hot

17. A hallway light can be turned on and off from both ends of the hall. Which switches are used?

A. Two SPST switches in series
B. Two SPST switches in parallel
C. One DPST switch
D. Two SPDT (3-way) switches

18. About what forward voltage does a silicon diode need before it conducts well?

A. 0.07 V
B. 0.7 V
C. 7 V
D. 70 V

19. On a new LED, the LONGER leg is the:

A. Cathode, connect it to -
B. Anode, connect it to +
C. Gate
D. It does not matter

20. A 2-input NAND gate has A = 1 and B = 1. The output is:

A. 0 **B.** 1

C. It depends on the last output **D.** Undefined

21. Which gates are called universal gates?

A. AND and OR **B.** XOR and XNOR

C. NAND and NOR **D.** NOT and buffer

22. By De Morgan's law, $(A \cdot B)'$ is equal to:

A. $A' \cdot B'$ **B.** $A + B$

C. $(A + B)'$ **D.** $A' + B'$

23. Before you measure a resistor with the ohmmeter setting of a multimeter, you must:

A. Remove the resistor from the powered circuit (power off)

B. Turn the circuit power on

C. Put the meter in series with the battery

D. Set the meter to the 10 A jack

Section II: Short answer and calculations

Write the answer with units and the correct number of significant figures. 2 points each.

24. Two small spheres carry +3.0 μC and +5.0 μC. Their centers are 0.20 m apart. Find the size of the electric force between them. Do they attract or repel? ($k = 8.99 \times 10^9$ N·m^2/C^2)

25. How many extra electrons give a small object a charge of −4.8 μC?

26. A US outlet is rated 120 V (rms). What is the peak voltage?

27. A 2400 mAh battery runs a toy that draws 300 mA. About how many hours does it last?

28. A lamp on 120 V draws 0.50 A. What is its power?

29. A toaster is rated 1100 W at 120 V. What is its resistance when it is on?

30. A 900 W microwave oven runs 30 minutes a day for 30 days. Electricity costs $0.13 per kWh. What is the cost?

31. A copper wire is 20.0 m long and 0.80 mm in diameter. Copper has $\rho = 1.68 \times 10^{-8}$ Ω·m. Find its resistance.

32. Find the equivalent resistance of 60 Ω, 30 Ω and 20 Ω in parallel.

33. A resistor has the bands red, red, orange, gold. Give its value and tolerance.

34. A 12.0 V source is across 4.7 kΩ (top) in series with 10 kΩ (bottom). What is the voltage across the 10 kΩ resistor?

35. A 120 V circuit has a 15 A fuse. What is the largest number of 65 W bulbs that can run at the same time without blowing the fuse?

36. A red LED (forward voltage 2.1 V) must run at 15 mA from a 6.0 V supply. What series resistor is needed?

37. Write the decimal number 45 in binary.

38. Add the binary numbers 1011 + 0111. Give the answer in binary.

39. Simplify F = AB + AB′.

40. A 2-input gate has the truth table below (A, B → F): 00 → 0, 01 → 1, 10 → 1, 11 → 0. Name the gate and write its Boolean expression.

41. A known 1.0 kΩ resistor is in series with a mystery resistor R_x. A voltmeter reads 2.4 V across the known resistor and 3.6 V across R_x. Find R_x.

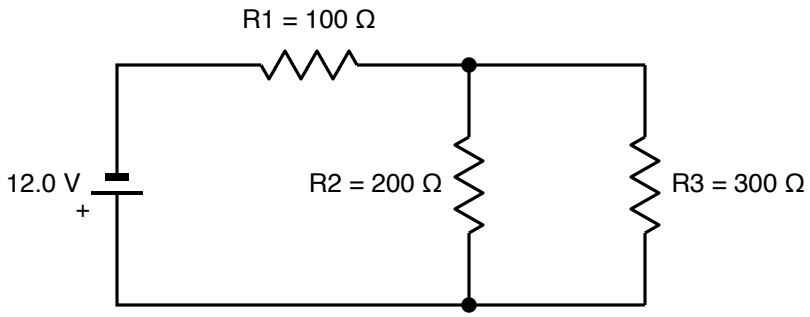
42. A phone battery is 3000 mAh at 3.7 V. The USB charger label says 5.0 V, 1.0 A. Ignore losses. Use the battery's ENERGY (Wh) and the charger's power. About how long does a full charge take?

Section III: Circuit problems

Show your work. 2 points per part.

Problem A: series-parallel circuit (Questions 43–47)

A 12.0 V battery drives R1 = 100 Ω in series with R2 = 200 Ω and R3 = 300 Ω, which are in parallel with each other (see the figure). Use ideal wires and an ideal battery.



43. Find the equivalent resistance of the whole circuit.

44. Find the current through the battery.

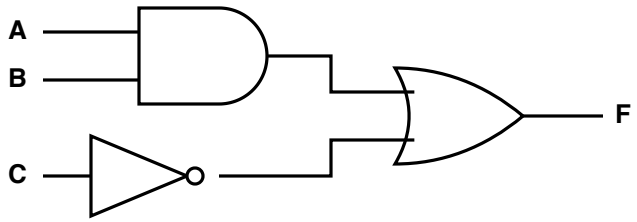
45. Find the voltage across R2.

46. Find the current through R3.

47. Find the total power that the battery supplies.

Problem B: logic circuit (Questions 48–50)

The logic circuit below has inputs A, B and C and output F.



48. Write the Boolean expression for F.

49. Find F when A = 1, B = 0, C = 0.

50. How many of the 8 input rows give F = 0? List them as ABC.

End of the test. Check your answer sheet.

Answer sheet · Practice Test 1 (Regional level)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

1.	**A**	**B**	**C**	**D**	**26.**	_____
2.	**A**	**B**	**C**	**D**	**27.**	_____
3.	**A**	**B**	**C**	**D**	**28.**	_____
4.	**A**	**B**	**C**	**D**	**29.**	_____
5.	**A**	**B**	**C**	**D**	**30.**	_____
6.	**A**	**B**	**C**	**D**	**31.**	_____
7.	**A**	**B**	**C**	**D**	**32.**	_____
8.	**A**	**B**	**C**	**D**	**33.**	_____
9.	**A**	**B**	**C**	**D**	**34.**	_____
10.	**A**	**B**	**C**	**D**	**35.**	_____
11.	**A**	**B**	**C**	**D**	**36.**	_____
12.	**A**	**B**	**C**	**D**	**37.**	_____
13.	**A**	**B**	**C**	**D**	**38.**	_____
14.	**A**	**B**	**C**	**D**	**39.**	_____
15.	**A**	**B**	**C**	**D**	**40.**	_____
16.	**A**	**B**	**C**	**D**	**41.**	_____
17.	**A**	**B**	**C**	**D**	**42.**	_____
18.	**A**	**B**	**C**	**D**	**43.**	_____
19.	**A**	**B**	**C**	**D**	**44.**	_____
20.	**A**	**B**	**C**	**D**	**45.**	_____
21.	**A**	**B**	**C**	**D**	**46.**	_____
22.	**A**	**B**	**C**	**D**	**47.**	_____
23.	**A**	**B**	**C**	**D**	**48.**	_____
24.	_____				**49.**	_____
25.	_____				**50.**	_____